INCH-POUND

MIL-DTL-32373B
1 March 2023
SUPERSEDING
MIL-DTL-32373A
12 August 2019

DETAIL SPECIFICATION

PLUG, MACHINE THREAD, MAGNETIC: DRAIN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

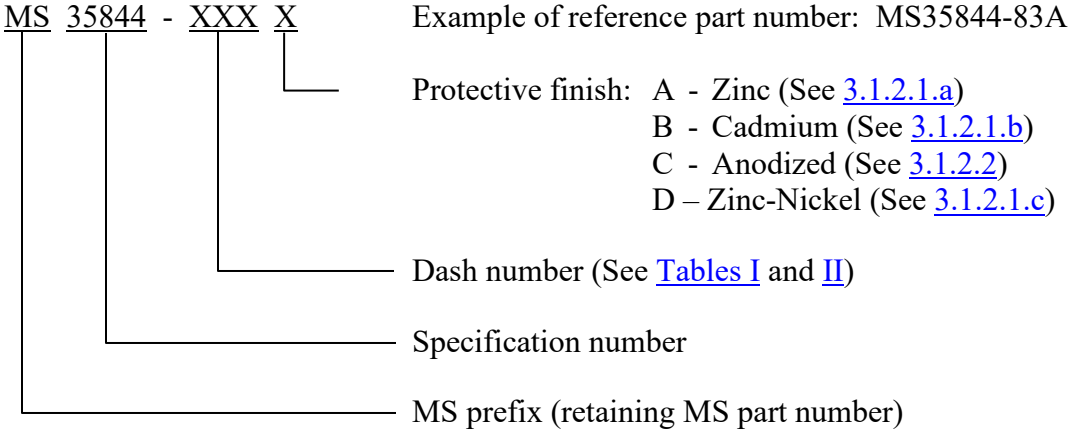
1. SCOPE

1.1 Scope. This specification covers two types of machine thread magnetic drain plugs used to seal engine oil lubrication systems and to capture ferrous contaminants in engine oil.

1.2 Classification. The machine thread magnetic drain plugs are of the following types as specified below:

Type I - External hexagonal (hex) head
Type II - External hex head with 3/4-inch square drive

1.3 Part or identifying number (PIN). The PIN to be used for drain plugs acquired to this specification is created as follows (see 6.2):



Comments, suggestions, or questions on this document should be addressed to DLA Aviation-VEB, 6090 Strathmore Rd. Richmond, VA 23237, or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.dla.mil/.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARD

FED-STD-H28 - Screw-Thread Standards for Federal Services

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-PRF-8625 - Anodic Coatings for Aluminum and Aluminum Alloy

DEPARTMENT OF DEFENSE STANDARD

MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-1916 - DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY

ASQ H1331 - Zero Acceptance Number Samplings Plans

(Copies of this document are available online at https://www.asme.org/)

ASME

ASME B46.1 - Surface Texture (Surface Roughness, Waviness, and Lay)

(Copies of this document are available online at https://www.asme.org.)

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ASTM INTERNATIONAL

ASTM A29/A29M	- Standard Specification for General Requirements for Steel Bars, Carbon and Alloy, Hot-Wrought[1].
ASTM A47/A47M	- Standard Specification for Ferritic Malleable Iron Castings
ASTM A108	- Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished
ASTM B211/B211M	- Standard Specification for Aluminum and Aluminum-Alloy Rolled or Cold Finished Bar, Rod, and Wire
ASTM B633	- Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel

(Copies of these documents are available online at https://www.astm.org.)

INTERNATIONAL ELECTROTECHNICAL COMMISSION

IEC 60404-1	- Magnetic materials – Part 1: Classification

(Copies of these documents are available online at https://www.iec.ch/.)

SAE INTERNATIONAL

SAE AMS-QQ-P-416	- Plating, Cadmium (Electrodeposited)
SAE AMS2461	- Plating, Zinc-Nickel Alloy (12 to 16% Ni)
SAE AS35769	- Gasket, Metallic, Encased, Annular, Copper
SAE J532	- Automotive Straight Thread Filler and Drain Plugs

(Copies of these documents are available online at https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials.

3.1.1 Machine thread drain plug body. Materials used in the manufacture of the machine thread drain plug body shall be as specified in 3.1.1.1 through 3.1.1.3.

3.1.1.1 Carbon steel. Carbon steel shall conform to compositional grade designations 1011 through 1025 in accordance with ASTM A29/A29M. Dimensional requirements for steel stock shall conform to ASTM A108.

3.1.1.2 Aluminum alloy. Aluminum alloy shall conform to chemical composition 2011, temper T3, or composition 2017, temper T4, in accordance with ASTM B211/B211M.

3.1.1.3 Malleable iron. Malleable iron shall conform to grade 32510 in accordance with ASTM A47/A47M.

3.1.2 Protective finishes.

3.1.2.1 Carbon steel and malleable iron. Drain plugs produced from carbon steel or malleable iron shall be one of the following:

a. Zinc[1/] plated in accordance with ASTM B633, class Fe/Zn 12, type II.
b. Cadmium[1/] plated in accordance with SAE AMS-QQ-P-416, type II, class 2; class 3 cadmium plating is permissible for thread sizes of 1/2-inch or less.
c. Zinc-nickel plated per SAE AMS2461, Type II, Grade A or B, Class 2.

[1/] Cadmium plating and Zinc plating as specified for PIN suffixes A and B contain cadmium and/or hexavalent chromium which are R1, R2, and D1 hazardous substances in the Aerospace and Defense Declarable Substance List. PIN suffix D, Zinc-Nickel plate per SAE AMS2461, Type II, Grade A or B, Class 2, is a suitable replacement with design authority approval.

3.1.2.2 Aluminum alloys. Drain plugs produced from aluminum alloys shall be anodized using type IC or IIB coatings in accordance with MIL-PRF-8625.Sealers containing hexavalent chromium shall be prohibited.

3.1.3 Magnet types. Magnets shall be of the permanent magnet type and materials shall be of the R1-1-1 class or the S1 class, any subclass conforming to IEC 60404-1.

3.2 Dimensions. Dimensions shall conform to those shown in figure 1 and tables I and II. Dimensions and features not described herein shall be in accordance with SAE J532.

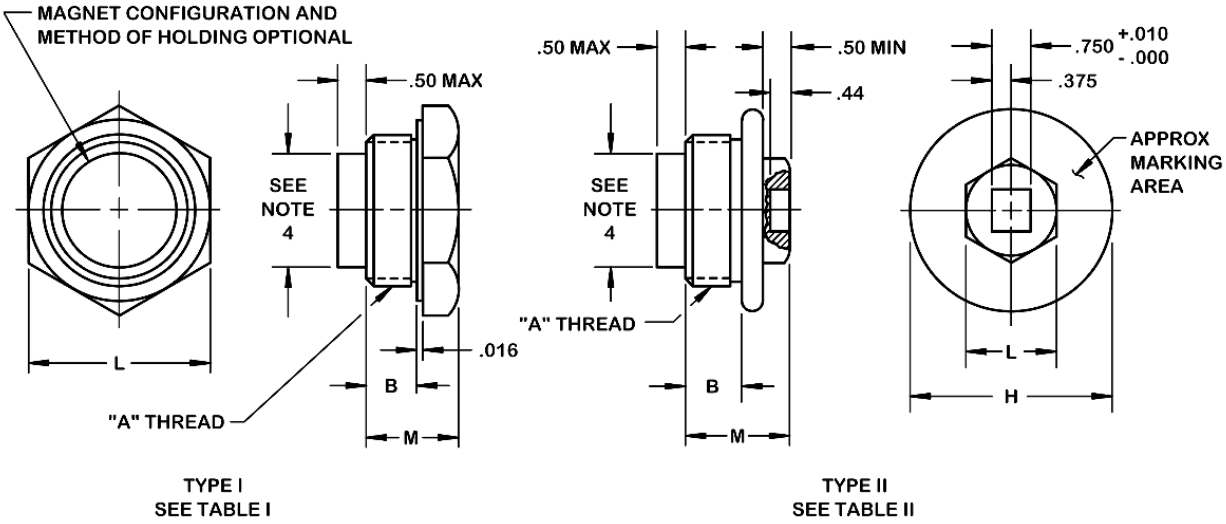
3.3 Magnets.

3.3.1 Magnetic lift. Magnets shall lift a minimum of 6 ounces of soft steel after being installed into the drain plug (see 4.4.2.1).

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3.3.2 Temperature. Magnets shall be firmly retained in the drain plug at temperatures up to 500 °F (260 °C) (see 4.4.2.2).

3.4 Threads. Threads shall be in accordance with FED-STD-H28 (see tables I and II).



NOTES:
1. All dimensions are in inches. Dimensions are shown in tables I and II. Unless otherwise specified, tolerances are ±0.010 on machined surfaces.
2. All machined surfaces shall have a surface texture not greater than 125 micrometers (µm) from the nominal; and all unmachined surfaces shall have a surface texture not greater than 500 µm from the nominal in accordance with ASME B46.1.
3. Across-corner dimensions of the hexagonal heads shall be a minimum of 1.092 times larger than the nominal across-flats width of the hexagonal heads (dimension "L").
4. The maximum magnet diameter shall be not less than 0.100 less than the thread pitch diameter.

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FIGURE 1. Machine thread magnetic drain plugs.

TABLE I. Type I machine thread magnetic drain plug dash numbers.

MS dash number (steel)	MS dash number (aluminum)	"A" thread size (Class 2A)	"B"	"L" (nominal)	"M"	Gasket dash number [1]
-1	-2	3/8(.375)-24 UNF	0.31	0.625	0.50	6
-49	-50	1/2(.50)-20 UNF	0.34	0.750	0.53	9
-3	-4	5/8(.625)-18 UNF	0.38	0.875	0.62	11
-83	-84	3/4(.75) -16 UNF	0.38	1.000	0.62	15
-5	-6	7/8(.875)-14 UNF	0.41	1.125	0.66	18
-99	-100	7/8(.875)-18 UNS	0.41	1.125	0.66	18
-111	-112	1(1.00)-18 UNS	0.44	1.250	0.75	21
-7	-8	1-1/4(1.25)-12 UNF	0.47	1.500	0.78	31
-133	-134	1-1/4(1.25)-18 UNEF	0.47	1.500	0.78	31
-149	-150	1-1/2(1.50)-18 UNEF	0.50	1.750	0.88	48

[1] In accordance with SAE AS35769 (see 3.5).

TABLE II. Type II machine thread magnetic drain plug dash numbers.

MS dash number (iron)	MS dash number (aluminum)	"A" thread size	"B"	"H" flange diameter	"L" (nominal)	"M"	Gasket dash number [1]
-9	-10	2-12 UN-2A	0.56	2.25	1.500	1.37	55
-167	-168	2-16 UN-2A	0.56	2.25	1.500	1.37	55
-11	-12	3-12 UN-2A	0.56	3.25	1.500	1.37	76

[1] In accordance with SAE AS35769 (see 3.5).

3.5 Gaskets. Gaskets shall conform to SAE AS35769 and shall be appropriately sized to the drain plug (see tables I and II).

3.6 Markings. Markings shall consist of the MS part number (see 1.3) and the manufacturer commercial and government entity (CAGE) code in accordance with MIL-STD-130 (see figure 1).

3.7 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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4. VERIFICATION

4.1 Classification of tests. All inspections required herein for the testing of drain plugs are classified as conformance tests and shall be conducted in accordance with the sampling techniques and methods of testing specified herein.

4.2 Sampling for inspection. A random sample of drain plugs for visual inspection and testing shall be selected from each lot in accordance with MIL-STD-1916, verification level (VL) IV or ASQ H1331, acceptance quality limit (AQL) 1.0 unless otherwise specified in the contract (see 6.2.c).

4.3 Lot. A lot shall consist of drain plugs of the same type, materials, and construction produced continuously by one machine or one series of progressive processing machines.

4.3.1 Rejection. The failure of any specimen shall be cause for rejection of the lot represented by the sample.

4.3.2 Certified test report. If specified (see 6.2), the manufacturer shall furnish a certified test report showing that the manufacturer's product satisfactorily conforms to this specification. The test report shall include, as a minimum, actual results of the tests specified herein.

4.4 Tests. Unless otherwise specified, all tests described herein shall be conducted at ambient room temperature.

4.4.1 Examination of product. All drain plug samples shall be examined for conformance to the requirements as shown in table III.

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TABLE III. Conformance test requirements.

Test requirement	Requirement paragraph	Requirement standard
Materials		
Machine thread magnetic drain plug		
Carbon steel - composition	3.1.1.1	ASTM A29/A29M
- stock dimensions	3.1.1.1	ASTM A108
Aluminum alloy	3.1.1.2	ASTM B211/B211M
Malleable iron	3.1.1.3	ASTM A47/A47M
Protective finish		
Zinc plating (steel or iron)	3.1.2.1	ASTM B633
Cadmium plating (steel or iron)	3.1.2.1	SAE AMS-QQ-P-416
Anodization (aluminum alloy)	3.1.2.2	MIL-PRF-8625
Zinc-Nickel plating (steel or iron)	3.1.2.1.	SAE-AMS2461
Magnets	3.1.3	IEC 60404-1
Dimensions	3.2	Figure 1, tables I and II
Machine tolerances	3.2	Figure 1
Surface texture	3.2	ASME B46.1
Plug threads	3.4	FED-STD-H28
Gasket	3.5	SAE AS35769

4.4.1.1 Rejects. Failure of any sample to meet the stated requirements shall be cause for rejection of the entire lot.

4.4.2 Magnets. Machine thread drain plugs with installed magnets shall be tested as described in 4.4.2.1 and 4.4.2.2.

4.4.2.1 Magnet lift. Using a drain plug with magnet installed, lift a solid piece of soft steel weighing not less than six ounces. Apply the magnet to the steel piece and manually raise the drain plug so that only the magnet lifts the steel piece. Hold the drain plug and steel in a suspended position for not less than five seconds. Failure of the magnet to hold the steel piece in suspension for five seconds shall be cause for rejection (see 3.3.1).

4.4.2.1.1 Rejects. Samples failing the magnet retention test shall result in rejection of the entire lot.

4.4.2.2 Magnet retention. Using an oven with thermostat control, heat the sample drain plug with installed magnet to 500 °F (260 °C). Once the test sample is at temperature, remove the sample from the oven using tongs and insulated gloves (Warning: The test sample will be very hot.). Use pliers to grasp and apply light torque to the magnet to verify that the magnet is firmly affixed to the drain plug. Do not overtorque. Failure of the magnet to remain firmly retained in the drain plug at temperature shall be cause for rejection (see 3.3.2).

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4.4.2.2.1 Rejects. Samples failing the magnet temperature test shall result in rejection of the entire lot.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The machine thread magnetic drain plugs covered by this specification are intended for use in engine oil lubrication systems as a means to seal and access the engine oil reservoir and to magnetically attract and capture ferrous contaminants in the engine oil that could result in accelerated engine wear.

6.1.1 Type I. Type I machine thread magnetic drain plugs feature an external hexagonal head that provides the surfaces against which torque is applied when loosening or tightening the drain plug (see figure 1).

6.1.2 Type II. Type II machine thread magnetic drain plugs feature a circular flange with an integral hexagonal head into which a 3/4-inch square drive slot has been incorporated (see figure 1). Either the hexagonal head or the square drive slot may be used to apply torque when loosening or tightening the drain plug.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. PIN (see 1.3).
c. AQL/VL (see 4.2).
d. Test report, if required (see 4.3.2).
e. Packaging requirements (see 5.1).

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6.3 Safety Data Sheet (SDS). If the use of cadmium is not considered an article as specified in OSHA 3844, "Hazard Communication: Hazardous Classification Guidance for Manufacturers, Importers, and Employers", a SDS must be prepared and submitted in accordance with FED-STD-313 (obtain from https://assist.dla.mil). The SDS must also meet the requirements of 29 CFR 1910.1200. The 29 CFR 1910.1200 requires that the SDS for each hazardous chemical used in an operation must be readily available to personnel using the material.

6.4 Subject term (key word) listing.

Cadmium
Contaminant
Engine wear
External head
Ferrous
Motor oil
Oil pan
Seal
Shaving
Square drive
Trap

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AT
Navy - YD
Air Force - 99
DLA - GS

Review Activities:
Army - CR4
Navy - MC, OS
DLA - CC

Preparing Activity:
DLA - GS4

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